ELISA BRUNI

SOIL ORGANIC CARBON MODELER

Paris @ebruni93 @elisabruni

WHO AM I

I hold a PhD in Soil Science.
My work focuses on the simulation of the soil organic carbon cycle with a multi-modeling approach.
I use calibration methods that enable the upscaling of model simulations from site to larger scales.

TECHNICAL SKILLS

Mathematical modeling of dynamic systems
Programming (R, python, Matlab, basic Fortran and Linux)

LANGUAGES

English
French
Italian
Spanish
Portuguese

WORKING EXPERIENCE

Post-doc | Holisoils
ENS (Ecole Normale Supérieure de Paris) | France
April 2022 - present
Multi-modeling approach to study the effect of forest management on soil organic carbon stocks and greenhouse gas emissions in Europe

Visiting research fellowship | FAPERJ
UFRRJ (Universidade Federal Rural de Rio de Janeiro) | Brazil
April 2022 - July 2022
Modeling the evolution of soil organic carbon stocks under conservation agriculture in tropical and subtropical climates

PhD | CLAND
LSCE (Laboratoire des Sciences du Climat et de l'Environnement) - Ecosys (INRAe) | France
December 2018 - March 2022
Soil organic carbon modeling: estimating carbon input changes required to reach policy objectives aimed at increasing soil organic carbon stocks

Master internship | CLUES
LSCE (Laboratoire des Sciences du Climat et de l'Environnement) | France
March 2018 - September 2018
Assessing the drivers behind leaf stoichiometric changes in European forests: a combined approach of statistical analyses and model simulations

PUBLICATIONS

Bruni, Elisa, Claire Chenu, Rose Z. Abramoff, Guido Baldoni, Dietmar Barkusky, Hugues Clivot, Yuanyuan Huang, et al. "Multi-modelling Predictions Show High Uncertainty of Required Carbon Input Changes to Reach a 4‰ Target." European Journal of Soil Science 73, no. 6 (November 2022). https://doi.org/10.1111/ejss.13330.

Bruni, E., Guenet, B., Clivot, H., Kätterer, T., Martin, M., Virto, I., and Chenu, C.: Defining Quantitative Targets for Topsoil Organic Carbon Stock Increase in European Croplands: Case Studies With Exogenous Organic Matter Inputs, Front. Environ. Sci., 10, 824724, https://doi.org/10.3389/fenvs.2022.824724, **2022**.

Bruni, E., Guenet, B., Huang, Y., Clivot, H., Virto, I., Farina, R., Kätterer, T., Ciais, P., Martin, M., and Chenu, C.: Additional carbon inputs to reach a 4 per 1000 objective in Europe: feasibility and projected impacts of climate change based on Century simulations of long-term arable experiments, Biogeosciences, 18, 3981–4004, https://doi.org/10.5194/bg-18-3981-2021, **2021**.

Guenet, B., Gabrielle, B., Chenu, C., Arrouays, D., Balesdent, J., Bernoux, M., **Bruni**, E., Caliman, J., Cardinael, R., Chen, S., Ciais, P., Desbois, D., Fouche, J., Frank, S., Henault, C., Lugato, E., Naipal, V., Nesme, T., Obersteiner, M., Pellerin, S., Powlson, D. S., Rasse, D. P., Rees, F., Soussana, J., Su, Y., Tian, H., Valin, H. and Zhou, F.: Can N 2 O emissions offset the benefits from soil organic carbon storage?, Glob. Change Biol., gcb.15342, doi:10.1111/gcb.15342, **2020**.